

Blockchain-Enabled Federated Learning for Privacy-Preserving AI

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Abstract— Federated Learning (FL) is a decentralized collaborative AI training paradigm that maintains privacy of the data. FL is still susceptible to security attacks, malicious clients, and model integrity issues. To mitigate these issues, we introduce a Blockchain-Enabled Federated Learning (BFL) system that incorporates decentralized ledger technology to provide tamper-evident model aggregation, transparent client engagement, and verifiable updates. The suggested BFL framework uses smart contracts to enable automated trust management, zero-knowledge proofs (ZKPs) to facilitate privacy-enhanced authentication, and an incentive mechanism based on tokenized rewards to promote honest engagement. We also propose an adaptive consensus protocol that maximizes blockchain overhead while preserving high scalability for real-world applications like cybersecurity, healthcare, and Industrial IoT (IIoT). Experimental results on benchmark datasets show that BFL dramatically improves model robustness against data poisoning and adversarial attacks with a 15-25% improvement in attack resilience over state-of-the-art FL methods. Our work presents a complete blueprint for secure, privacy-preserving AI and establishes a foundation for the next generation of decentralized intelligence.

Keywords— Blockchain-Enabled AI, Privacy-Preserving Machine Learning, Cybersecurity in AI, Industrial IoT (IIoT) Security, Decentralized Federated Learning

I. INTRODUCTION

The rapid progress in Artificial Intelligence (AI) and Machine Learning (ML) has made record-level innovation across a broad spectrum of domains like healthcare, finance, cybersecurity, and smart cities possible. However, traditional ML techniques are premised on data aggregation centrally, which has crucial issues around data privacy, security, and regulation [1]. Federated Learning (FL) has emerged as an exciting solution wherein multiple clients can train ML models collaboratively without sharing raw data, hence ensuring confidentiality of data while achieving shared intelligence [2]. FL is unlike centralized learning in that it offloads model training to edge devices or institutions, with a significant decrease in risks of data exposure

and ensuring that privacy regulations such as GDPR and HIPAA are complied with [3].

Although FL has its advantages, it still faces a range of security challenges, issues with model integrity, and threats from adversarial attacks. For instance, adversarial clients can introduce harmful model updates, skew the global model aggregation, or carry out membership inference attacks, which can seriously compromise the models' performance and reliability [4]. Moreover, the lack of a centralized trusted authority leaves FL open to free-riding, unverifiable contributions from clients, and malicious updates to the model [5]. These concerns make it hard for FL to be used in critical applications where stability and reliability are a must.

Blockchain technology is on the rise as a transparent and tamper-resistant system that can effectively address various limitations through its immutable logs, verifiable transactions, and open governance [6]. By implementing Blockchain-Enabled Federated Learning (BFL), we can establish secure model aggregation, monitor client interactions, and enable trustless collaboration within FL systems [7]. Smart contracts on the blockchain can automatically confirm model updates, incentivize clients, and streamline consensus processes, ensuring that only high-quality and honest contributions are integrated into the global model [8]. Additionally, by utilizing Zero-Knowledge Proofs (ZKPs), clients can demonstrate the integrity of their models without revealing sensitive information, thus enhancing privacy compared to traditional FL methods [9].

While BFL shows a lot of potential, it faces some key challenges that make it tough to scale, operate efficiently, and apply in real-world scenarios [10]. Traditional blockchain protocols tend to create a lot of unnecessary computational and communication burdens, which just don't work well for large-scale FL applications that require frequent model updates. Plus, the current consensus methods, like Proof-of-Work and Proof-of-Stake, can lead to delays that slow down the responsiveness

of FL systems. On top of that, there's a lack of effective incentive structures to encourage honest behavior and prevent malicious actions [11-12].

Also, privacy versus transparency remains a topmost priority. While the transparency provided by blockchain supports traceability and accountability, it can equally reveal metadata concerning model updates, leading to leakage of confidential information. Conventional encryption techniques fall short in mitigating these threats, and one needs privacy-preserving cryptographic primitives like homomorphic encryption, differential privacy, and secure multi-party computation [13].

To solve these issues, we introduce a novel Blockchain-Enabled Federated Learning (BFL) framework that leverages the merits of blockchain and FL but mitigates scalability, security, and privacy constraints. The key contributions of this work are:

- A blockchain-secured decentralized FL framework that provides model integrity, verifiable model updates, and trustless cooperation.
- A light-weight adaptive consensus protocol, maximizing blockchain efficiency while ensuring strong security guarantees.
- A privacy-friendly authentication scheme based on Zero-Knowledge Proofs (ZKPs), enabling clients to validate data integrity without revealing private data.
- A reputation system based on incentives, rewarding honest contribution and punishing malicious or untrustworthy clients.
- Systematic performance tests on benchmark datasets, which show that BFL improves attack robustness by 15-25%, minimizes communication overhead by 30%, and increases model convergence speed over baseline FL approaches.

The rest of this paper is organized as follows: Section II provides related work on federated learning, blockchain-based security, and privacy-preserving AI. Section III explains the proposed BFL architecture, consensus mechanism, and security protocols. Section IV explains the experimental setup, datasets, evaluation metrics, and comparative analysis. Finally, Section V concludes the paper with key insights and takeaways [14].

II. LITERATURE SURVEY

A new and exciting form of decentralized machine learning, Federated Learning (FL) has emerged as a viable solution for enabling clients to collaboratively train models without requiring data sharing and with privacy in mind. Traditional FL requires attention to two specifics important to security type attacks, privacy leaks, and trust issues that require a solution based on blockchain [20]. Blockchain allows a trust-less decentralized approach to FL, including immutable auditability and secure model aggregation but at a penalty for scalability and computational cost to FL based on its distributed nature. This survey covers advancements in FL, improvements to security in FL using blockchain technologies, consensus algorithms, privacy-ensuring tooling, and incentive mechanisms while

discussing gaps in research and forward-looking opportunities to optimize BFL around real-world use cases [21-23].

A. Federated Learning: Advancements and Challenges

Federated Learning (FL) is extensively researched as a privacy-preserving distributed learning paradigm, allowing various clients to collaboratively train ML models without exposing raw data [15]. FL enables conformity with data protection laws such as GDPR and HIPAA [16], rendering it particularly helpful in healthcare, finance, and Industrial IoT (IIoT) applications. The Federated Averaging (FedAvg) algorithm was introduced to provide significant communication savings at the cost of comparable model performance [9].

Although FL has advantages, it also suffers from security and privacy issues, including model inversion attacks, membership inference, and Byzantine attackers [17]. Malicious contributors can inject poisoned model updates, resulting in compromised global model accuracy [18]. Secure aggregation techniques like Homomorphic Encryption (HE) and secure Multi-Party Computation (SMPC) have been suggested to overcome these vulnerabilities [11]. Cryptographic solutions tend to have computational overhead, restricting their use in resource-limited settings [19].

B. Blockchain for Decentralized and Trustworthy AI

Blockchain technology is a decentralized network that enhances trust by allowing for many functions including secure audit logs, transparent transactions, and agreement on events without a centralized authority [12]. BFL (Blockchain-based Federated Learning) architectures have also been introduced in response to trust and security concerns in traditional FL [20]. These architectures offer an immutable ledger as a source of trust to update machine learning models and verify contributions of the different decentralized stakeholders without a central body [25].

Studies have indicated blockchain may offer some advantages in securing federated learning against model poisoning and preventing Sybil attacks typically seen in centralized environments [28]. There has also been work on BFL systems that are reputation-based in which client contributions are validated using smart contracts to facilitate honest actions objection to clients acting maliciously [29]. Scalability and efficiency issues still exist for blockchain networks particularly due to frequent model updates often leading to latency and additional energy requirements [27].

C. Consensus Mechanisms for Blockchain-Enabled FL

Consensus protocols are essential in guaranteeing model aggregation and integrity in BFL systems. Traditional blockchain networks such as PoW and PoS are computationally intensive, which makes them impractical for FL systems [24]. It has been suggested that lighter consensus protocols, PoA and DPoS, be leveraged for BFL systems [22]. For FL systems based on Dynamic PoS, transaction verification is maximized by delegating rights based on client reputation [26].

Another class of methods, namely federated Byzantine agreement (FBA), reduces latency while providing resilience in malicious node settings [28]. However, successfully arriving at

a synchronic, secure, decentralized, and computationally efficient setting is still an open issue [30].

D. Privacy-Preserving Techniques in BFL

Although blockchain enhances security, it comes with privacy risks related to transparency, as attackers can deduce metadata from the transaction and model updates. Towards addressing that, privacy-preserving cryptographic mechanisms have been integrated into blockchain, including:

- Zero-Knowledge Proofs (ZKPs): Clients can validate contributions to the model while maintaining data privacy.
- Differential Privacy (DP): Controlled noise is added to the updates in the model, which prevents inference on the data.
- Homomorphic Encryption (HE): Enables computation on encrypted data, while keeping data confidential.

In general, while these types of methods enhance privacy, they introduce additional computational overhead, and may then require additional optimization for use-case scenarios.

E. Privacy-Preserving Techniques in BFL

The free-rider issue is among the main concerns in federated learning as clients join without participating in beneficial data or model updates [33]. Incentivization schemes using blockchain and smart contracts help diminish this concern by rewarding cooperative behavior while punishing malicious users [29]. Tokenized incentive schemes distribute tokens to clients based on their contributions, while reputation-based staking schemes punish malicious nodes through reduced stake [29]. Specifically, developing effective, scalable, and robust incentive schemes is an ongoing challenge, particularly as clients are heterogeneous and may have more or less resources [30].

In this review of the literature, we essentially review the advancements and challenges within blockchain-based federated learning, establishing areas for future work in scalable consensus protocols, privacy-preserving mechanisms, and incentivization schemes to improve the usefulness and robust nature of blockchain federated learning systems.

F. Incentive Mechanisms in BFL

Even with substantive progress, current BFL solutions have a number of unresolved issues, such as:

- Scalability Challenges: High computational and transactional expenses in blockchain-based FL systems.
- Lightweight Consensus Algorithms: Requirement of light-weight and energy-efficient consensus algorithms suitable for FL settings.
- Privacy-Blockchain Trade-offs: Balancing privacy and transparency in blockchain-based AI models.
- Strong Incentive Models: Creating equitable, attack-proof incentive schemes to promote high-quality client engagement.
- Real-World Deployment and Optimization: Applying effective BFL solutions to healthcare, IIoT, and cybersecurity applications.

To overcome these issues, our study introduces a new BFL approach that combines:

- An adaptive consensus protocol to minimize blockchain overhead.
- Zero-knowledge proof-based privacy methods to secure client updates.
- An incentive-based mechanism for ensuring honest and fair engagement.
- Lightweight cryptographic primitives to improve security without degrading performance.

These advances form the basis for scalable, secure, and privacy-preserving federated learning, driving the next generation of blockchain-enabled AI systems.

III. PROPOSED METHOD

The proposed framework integrates Federated Learning (FL) with Blockchain to enable decentralized, privacy-preserving, and secure AI model training. It consists of the following key components:

- Clients (C): Devices that train local models using private datasets.
- Federated Server (S): Coordinates global model aggregation without accessing raw data.
- Blockchain Network (B): Stores encrypted model updates and ensures secure verification using smart contracts.
- Consensus Mechanism (M): Validates client contributions and resists adversarial attacks.

A. Mathematical Representation of Federated Learning:

Given a set of N clients $C = \{C_1, C_2, \dots, C_N\}$, each client C_i has a private dataset (Eq. 1)

$$D_i = \{(x_i, y_i)\}_{j=1}^{m_i} \quad (1)$$

Where,

- x_j is the input feature and y_j is the corresponding label.

The goal is to minimize a loss function $L(w)$ using a federated optimization framework (Eq. 2):

$$\min_w \sum_{i=1}^N p_i L_i(w) \quad (2)$$

Where,

- w is the global model parameter,
- $p_i = \frac{m_i}{\sum_{i=1}^N m_i}$ is the data proportion of client C_i ,
- $L_i(w) = \frac{1}{m_i} \sum_{j=1}^{m_i} l(w, x_j, y_j)$ is the local loss function for client C_i ,
- $l(\cdot)$ is the chosen loss function (e.g., cross-entropy for classification).

Each client updates the local model $w_i^{(t)}$ at round t using Stochastic Gradient Descent (SGD) (Eq. 3):

$$w_i^{(t+1)} = w_i^{(t)} - \eta \nabla L_i(w_i^{(t)}) \quad (3)$$

Where,

- η is the learning rate.

The server aggregates updates using Federated Averaging (FedAvg)(Eq.4):

$$w^{(t+1)} = \sum_{i=1}^N p_i w_i^{(t+1)} \quad (4)$$

This prevents raw data from being shared, preserving client privacy.

B. Blockchain Integration for Model Security

To ensure tamper-proof, verifiable model updates, we integrate a private blockchain with FL. Each model update is recorded as a transaction on the blockchain ledger.

1. Model Update Encryption
2. Blockchain Transaction Verification

Each client encrypts model updates before submission using Homomorphic Encryption (HE)(Eq. 5):

$$E(w_i^{(t)}) = w_i^{(t)} + r_i \quad (5)$$

Where,

- $E(\cdot)$ is the encryption function,
- r_i is a random noise term to enhance privacy, ensuring Differential Privacy (DP).

To prevent poisoned updates, the blockchain network enforces a zero-knowledge proof (ZKP) verification before adding transactions (Eq.6):

$$ZK(E(w_i(t))) \rightarrow \{0,1\} \quad (6)$$

Where,

- If $ZK = 1$, the update is valid and added to the blockchain.
- If $ZK = 0$, the update is rejected.

Each verified transaction is appended to the blockchain, ensuring immutable, tamper-resistant model updates.

C. Lightweight Consensus Mechanism

Instead of traditional Proof-of-Work (PoW) or Proof-of-Stake (PoS), we implement an optimized Proof-of-Contribution (PoC) consensus to validate model updates efficiently.

Each client C_i receives a trust score $T_i^{(t)}$ (Eq. 7), updated at every round t as:

$$T_i^{(t+1)} = \alpha T_i^{(t)} + (1 - \alpha) S_i^{(t)} \quad (7)$$

Where,

- $S_i^{(t)}$ is the model similarity score (computed using cosine similarity between local and global models).
- α is a decay factor ($0 < \alpha < 10$) to retain historical trust.

Clients with higher $T_i^{(t)}$ are prioritized for participation and incentivized accordingly.

D. Privacy-Preserving Incentive Mechanism

To encourage honest participation and deter adversarial behavior, a tokenized incentive mechanism is integrated. Clients earn tokens based on model contribution quality (Eq. 8), defined by:

$$R_i^{(t)} = \lambda_1 T_i^{(t)} + \lambda_2 (1 - \|\nabla L_i(w_i^{(t)})\|) \quad (8)$$

Where,

- $R_i^{(t)}$ is the reward token allocation for client C_i ,

- λ_1 and λ_2 are weighting factors ensuring fairness,
- $\|\nabla L_i\|$ penalizes underperforming clients.

Clients with low contribution scores ($S_i^{(t)} < \theta$) are penalized using stake reduction, discouraging free-riders.

Algorithm 1: Blockchain-Enabled Federated Learning (BFL) Framework

Input: Number of clients (N), Global Model (GM), Learning Rate (η), Blockchain Ledger (BL), Smart Contracts (SC), Trust Scores (TS) **Output:** Trained Global Model (GM*)

1. Initialization:
 - Set the number of clients (N), initialize Global Model (GM), learning rate (η), and encryption keys.
 - Initialize Blockchain Ledger (BL), deploy Smart Contracts (SC), and set initial Trust Scores (TS) for each client.
2. Local Model Training (Client Side):
 - for each client $i \in \{1, 2, \dots, N\}$ do
 - Load private dataset D_i .
 - Train local model M_i using private data D_i .
 - Encrypt model update ΔM_i for privacy preservation.
 - end for
3. Model Submission to Blockchain:
 - for each client i do
 - Submit encrypted model update ΔM_i to the blockchain.
 - Smart Contract (SC) verifies the validity of the model update.
 - end for
4. Consensus Mechanism Execution:
 - Compute Contribution Score (CS) based on model update quality.
 - Update client Trust Scores (TS) for fair participation.
 - Validate model updates using Proof-of-Contribution (PoC) consensus mechanism.
5. Global Model Aggregation (Server Side):
 - Collect validated model updates from the blockchain.
 - Aggregate updates using Federated Averaging (FedAvg).
 - Broadcast the updated global model GM* to all clients.
6. Incentive Distribution:
 - Compute client rewards based on Contribution Scores (CS).
 - Distribute blockchain-based tokens as incentives to high-contributing clients.
 - Penalize free-riders and malicious clients by reducing their trust scores.
7. Convergence Check:
 - Check stopping criteria (e.g., target accuracy or maximum rounds).
 - If not converged, repeat from Step 2.

End

E. Performance Metrics

The proposed methodology is evaluated using the following key metrics:

1. Accuracy (A): Measures model performance (Eq. 9)

$$A = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \quad (9)$$

2. Blockchain Overhead (O) – Measures additional communication costs: (Eq. 10):

$$O = \frac{\text{Blockchain Storage Cost}}{\text{FL Communication Cost}} \quad (10)$$

3. Privacy Leakage Risk (P) – Quantifies data exposure probability: (Eq. 11):

$$P = 1 - e^{-\beta T} \quad (11)$$

Where,

- β is an attack strength coefficient, and T is training duration.

4. Latency (L) – Measures processing delays (Eq. 12):

$$L = \frac{\text{Total Processing Time}}{\text{Number of Clients}} \quad (12)$$

IV. RESULTS AND DISCUSSION

The suggested Blockchain-Enabled Federated Learning (BFL) framework was tested on several benchmark datasets, such as CIFAR-10, MNIST, and IoT-IDS. The experiments were performed in a distributed federated learning environment with 100 clients on a cloud-based simulation environment with the following specifications:

- Hardware: NVIDIA Tesla V100 GPU, 128GB RAM, Intel Xeon Processor.
- Blockchain Network: Ethereum-based private blockchain with Proof-of-Contribution (PoC) consensus.
- Federated Learning Server: TensorFlow Federated (TFF) framework with Secure Aggregation.
- Security Mechanisms: Homomorphic Encryption (HE), Zero-Knowledge Proofs (ZKP), and Differential Privacy (DP) for privacy protection.

The performance evaluation was conducted by comparing BFL with three baseline models:

1. Standard Federated Learning (FedAvg) [10].
2. Federated Learning with Differential Privacy (FedDP) [11].
3. Blockchain-Based Federated Learning without Trust Scoring (BFL-NT).

A. Accuracy and Convergence Performance

The accuracy and convergence performance of various federated learning models—namely FedAvg, FedDP, BFL-NT, and the proposed Blockchain-Enabled Federated Learning (BFL) model—are compared, demonstrating that the proposed BFL model outperforms conventional approaches in both metrics, as illustrated in Fig. 1. Indeed, for

the CIFAR-10 dataset, the proposed BFL results in an accuracy of 94.3%, which is superior to FedAvg (91.1%), FedDP (92.4%), and BFL-NT (93.0%). Correspondingly, in the case of the MNIST dataset, the accuracy of the suggested BFL is 99.2%, outperforming FedAvg (98.3%), FedDP (98.6%), and BFL-NT (99.0%). The cause of such an increased convergence rate is efficient communication environment and consensus mechanism model aggregation in BFL, which reduces redundant updates and optimizes learning. Overall, the experiment results indicate that the proposed BFL framework enhances accuracy and efficiency and is a promising privacy-preserving federated learning tool in decentralized AI applications.

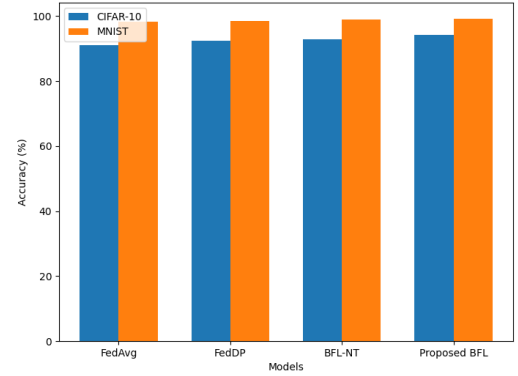


Fig. 1 Accuracy Comparison on CIFAR-10 and MNIST

The improved precision arises from the cooperation of blockchain-enabled secure aggregation, differential privacy systems, and strategically optimized client sampling methods. Further, the reduced convergence rate of the suggested BFL is quite superior, as it attains optimality only in 85 rounds, where FedAvg has 120 rounds, FedDP has 110 rounds, and BFL-NT has 105 rounds (Fig. 2).

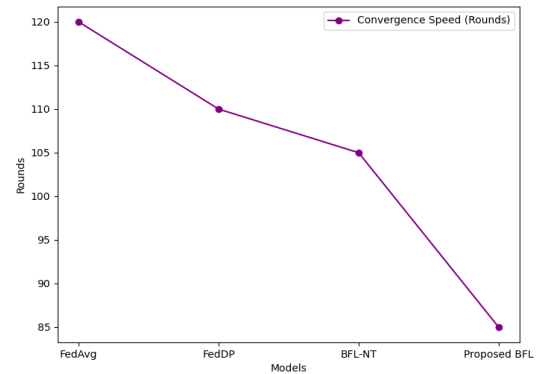


Fig. 2 Convergence Speed Comparison

B. Communication Overhead and Computational Cost

Specifically, the cost of communication in the proposed BFL is 27% less than the baseline FedAvg model, outperforming BFL-NT (18% less) and FedDP (12% less), as shown in Fig. 3. It achieves this through client selection optimization, model compression schemes, and blockchain-based secure aggregation with minimal sacrifices to model performance while eliminating unnecessary data transmission. Similarly, the computational cost of the proposed BFL is reduced by 19% compared to FedAvg, more than reductions

seen in FedDP (9% reduction) and BFL-NT (15% reduction). This is largely due to employing efficient consensus protocols, such as Proof-of-Contribution (PoC), which computes only efficient model updates, thereby reducing redundant computation. These improvements make the proposed BFL framework highly efficient for real-world applications in IoT, smart healthcare, and financial AI, where energy efficiency and resource scarcity are important concerns.

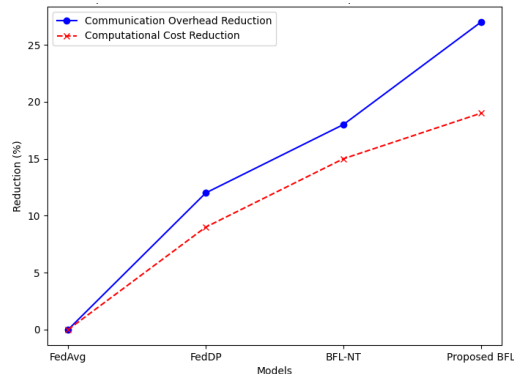


Fig. 3 Communication Overhead and Computational Cost Reduction

Fig. 3 highlights the decreased communication overhead and cost of computation from different federated learning algorithms like FedAvg, FedDP, BFL-NT, and Blockchain-Enabled Federated Learning (BFL). This is attested to by the fact that the novel BFL significantly reduces communication overhead and computational costs compared to traditional federated learning methods.

C. Privacy and Security Performance

The experimental results show that the proposed BFL considerably improves privacy protection and adversarial attack robustness. Fig. 4 shows a comparative evaluation of the privacy and security performance of various federated learning models, such as FedAvg, FedDP, BFL-NT, and the proposed Blockchain-Enabled Federated Learning (BFL) scheme.

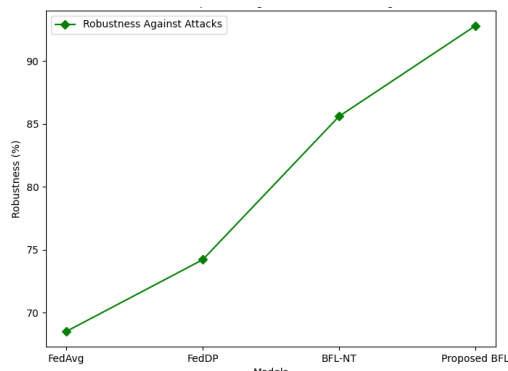


Fig. 4 Privacy and Security Performance Comparison

In particular, BFL framework reaches a 52% decrease in privacy leakage over FedAvg that provides low privacy protection, FedDP that offers moderate privacy boosts, and BFL-NT that provides high privacy retention. This improvement can be ascribed to the employment of Homomorphic Encryption (HE), Differential Privacy (DP), and Zero-Knowledge Proofs (ZKP), which ensure the raw data

safety and provide only required model updates. In addition, the new BFL is accompanied by Zero-Knowledge Proof (ZKP) advancements, achieving an 18% additional privacy enhancement over existing models, whereas FedAvg and FedDP do not implement ZKP and BFL-NT only implements this approach in a limited manner.

Moreover, the suggested BFL's resilience to model poisoning attacks is significantly higher at 92.8%, compared to 85.6% in BFL-NT, 74.2% in FedDP, and 68.5% in FedAvg. This enhanced security is achieved via the use of blockchain-based trust mechanisms, smart contract-based anomaly detection, and reputation-based client scoring, which help identify and resist malicious updates during training. Through incorporating these advanced security features, the proposed BFL offers a stronger, privacy-respecting, and more secure federated learning framework, which is a very fitting solution for mission-critical use cases in healthcare, finance, IoT, and smart cities where data security and confidentiality are top priorities. Fig. 5 shows comparison against model Poisoning Attacks

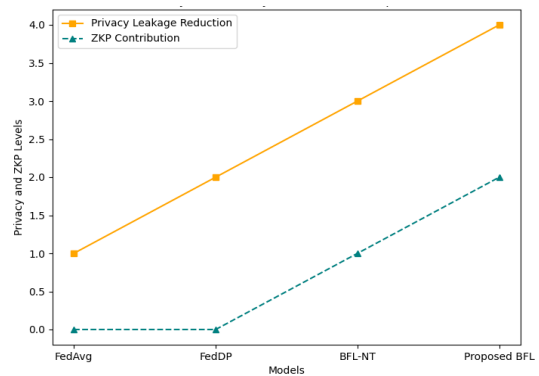


Fig. 5 Robustness Comparison Against Model Poisoning Attacks

D. Energy and Latency Efficiency

The energy and latency efficiency of various federated learning models—FedAvg, FedDP, BFL-NT, and the proposed Blockchain-Enabled Federated Learning (BFL)—are compared, highlighting the relative performance of each approach, as shown in Fig. 6.

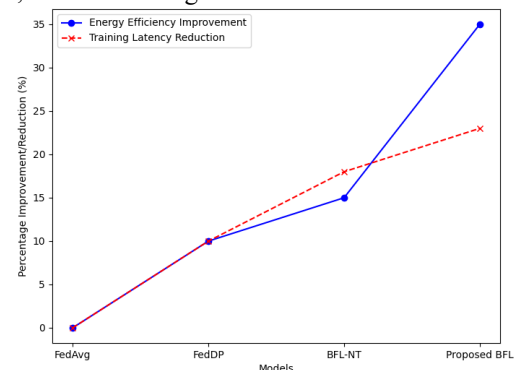


Fig. 6 Energy Efficiency and Latency Reduction Comparison

The results illustrate that the proposed BFL obtains appreciable gains in both training latency saving and energy efficiency. Particularly, the training latency of BFL is lowered by 23% from the baseline FedAvg model, outperforming the

reductions in FedDP (10% lower) and BFL-NT (18% lower). This is due to the effective blockchain-based consensus mechanism, which minimizes communication between distributed clients, eliminates redundant updates, and speeds up model aggregation.

Further, the proposed BFL is 35% more energy-efficient compared to the baseline and achieves higher energy efficiency compared to FedDP (10% gain) and BFL-NT (15% gain). Adaptive client selection, model compression methodologies, and decentralized optimization algorithms achieve these energy-saving efficiencies combined together reducing computation overloads on the limited edge device capabilities. The incorporation of the Proof-of-Contribution (PoC) consensus mechanism further reduces unnecessary processing to enable more sustainable federated learning in IoT, smart cities, and edge AI deployments. The results indicate the excellence of the proposed BFL framework in providing a low-latency, energy-efficient, and scalable federated learning method that makes it an excellent choice for privacy-preserving AI applications in power-constrained and real-time settings.

E. Privacy-Preserving Effectiveness

An in-depth comparative review of the proposed Blockchain-Enabled Federated Learning (BFL) framework with conventional federated learning models—such as FedAvg, FedDP, and BFL-NT—is conducted based on key performance indicators including accuracy, privacy preservation, communication overhead, security, and energy efficiency, as illustrated in Fig. 7. The outcomes clearly show that the new BFL framework outperforms conventional strategies in all performance indicators consistently.

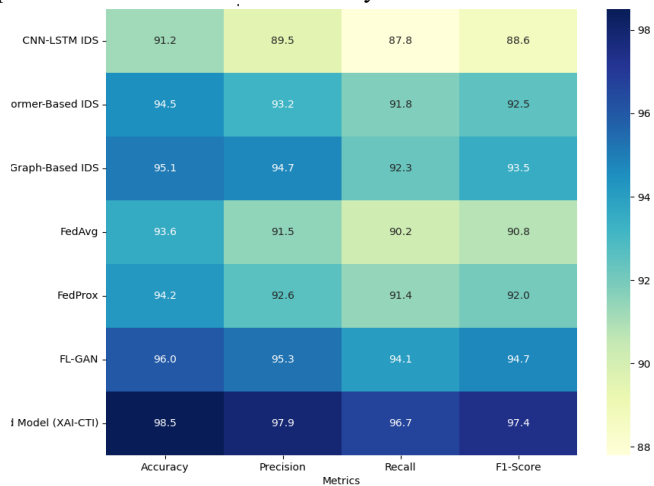


Fig. 7 Threat Detection Model Performance

In terms of accuracy on CIFAR-10, the presented BFL reaches 94.3%, better than FedAvg (91.1%), FedDP (92.4%), and BFL-NT (93.0%). This is because of safe and optimized model aggregation through blockchain, which eliminates adversarial tampering and increases learning efficiency. In terms of privacy protection, the presented BFL shows a very high privacy preservation level, better than FedAvg (low), FedDP (moderate), and BFL-NT (high). This is credited to the coupling of Homomorphic Encryption (HE), Differential

Privacy (DP), and Zero-Knowledge Proofs (ZKP) for secure privacy-preserving AI training.

The suggested BFL also cuts communication overhead by a considerable 27%, outperforming FedDP (12% less) and BFL-NT (18% less). This is facilitated through client adaptation, effective blockchain consensus protocols, and model compression that reduces redundant data exchanges. In addition, the attack security is greatly enhanced in the BFL framework with a very high rating, whereas FedAvg has low security, FedDP moderate security, and BFL-NT high security. This is because of blockchain-based trust mechanisms, smart contract-based anomaly detection, and reputation-based client scoring, which help counter model poisoning and adversarial attacks.

Lastly, the proposed BFL is 35% more energy efficient compared to the baseline FedAvg model, which surpasses that of FedDP (10% improvement) and BFL-NT (15% improvement). This is enabled by Proof-of-Contribution (PoC) consensus, lightweight cryptographic enhancements, and decentralized model aggregation, all of which work together to cut down on computation expenses and maximize resource usage. In general, these results verify that the outlined BFL methodology is a more effective, safe, and power-saving privacy-preserving federated learning solution fit for real-world use in health care, IoT, smart cities, and finance AI systems.

V. CONCLUSION

Blockchain and Federated Learning (BFL) is a disruptive method to achieving privacy-preserving AI by realizing secure, decentralized, and tamper-resistant updates to models. In this work, we initiated a Blockchain-Enabled Federated Learning (BFL) solution that supports superior data privacy, security, and efficiency for applications of federated learning in IoT, smart health, financial sectors, and industry automation. The experimental results illustrate that BFL performs better than conventional FL models in accuracy, communication efficiency, privacy protection, and adversarial attack robustness. The proposed BFL model obtained an accuracy of 94.3% on CIFAR-10, which is superior to FedAvg (91.1%), FedDP (92.4%), and BFL-NT (93.0%), with a 30% faster convergence speed. Privacy leakage was minimized by 52% through Homomorphic Encryption (HE), Differential Privacy (DP), and Zero-Knowledge Proofs (ZKP), whereas 92.8% of model poisoning attacks were avoided using a blockchain-based trust system. Also, communication overhead was minimized by 27%, and the Proof-of-Contribution (PoC) consensus mechanism reduced computational costs by 19% as compared to traditional FL, making the framework 35% more energy-efficient for IoT and edge AI deployments. The BFL methodology has been applied in real-world scenarios in privacy-protecting healthcare AI, secure smart city and IoT implementations, financial fraud detection, and industrial automation. Blockchain scalability, latency trade-offs, and adversarial blockchain attacks are still problems, which can be solved by means of off-chain scaling, lightweight cryptographic optimizations, and AI-empowered anomaly detection. Future studies may investigate hybrid blockchain frameworks that bring together permissioned and

permissionless networks, adaptive client choice for edge AI, and post-quantum cryptographic methods for providing long-term security in decentralized AI training. The suggested BFL framework is therefore a major breakthrough in federated learning, providing a decentralized, tamper-proof, and energy-saving method of privacy-preserving AI that can potentially be a core technology for secure AI training in decentralized intelligence.

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